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React Transitions

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What is useTransition?

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The `useTransition` hook helps you keep your React app responsive during heavy updates.

It lets you mark some state updates as "non-urgent", allowing other, more urgent updates to happen first.

When to Use Transitions?

Use transitions when you have:

- A slow operation that might freeze the UI
- Updates that aren't immediately critical
- Search results that take time to display

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Example

Get your own React.js Server

```
import { useState, useTransition } from 'react';

function SearchBar() {
  const [text, setText] = useState('');
  const [results, setResults] = useState('');
  const [isPending, startTransition] = useTransition();

  const handleChange = (e) => {
    // Urgent: Update input right away
    setText(e.target.value);

    // Non-urgent: Update search results
    startTransition(() => {
      setResults(e.target.value);
    });
  };

  return (
    <div>
      <input value={text} onChange={handleChange} />
      {isPending ? (
        <p>Loading...</p>
      ) : (
        <p>Search results for: {results}</p>
      )}
    </div>
  );
}
```

Run Example »

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- The loading message shows while the transition is pending

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Real-World Example

Here's a more practical example with a slow search feature:

Example

```
import { useState, useTransition } from 'react';

function SearchResults({ query }) {
  // Simulate slow search results
  const items = [];
  if (query) {
    for (let i = 0; i < 1000; i++) {
      items.push(<li key={i}>Result for {query} - {i}</li>);
    }
```

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```
function App() {
  const [input, setInput] = useState('');
  const [query, setQuery] = useState('');
  const [isPending, startTransition] = useTransition();

  const handleChange = (e) => {
    // Urgent: Update input field
    setInput(e.target.value);

    // Non-urgent: Update search results
    startTransition(() => {
      setQuery(e.target.value);
    });
  };

  return (
    <div>
      <input
        type="text"
        value={input}
        onChange={handleChange}
        placeholder="Type to search..."
      />
      {isPending && <p>Loading results...</p>}
      <SearchResults query={query} />
    </div>
  );
}
```

Run Example »

How this example works:

1. When you type in the input field, it updates immediately
2. The search results update is wrapped in `startTransition`

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useTransition Hook

The `useTransition` hook returns two items:

- `isPending` : tells you if a transition is active
- `startTransition` : function to mark updates as transitions

Note: Use transitions only for non-urgent updates. For example, typing in an input field should be urgent, but filtering a large list can be a transition.

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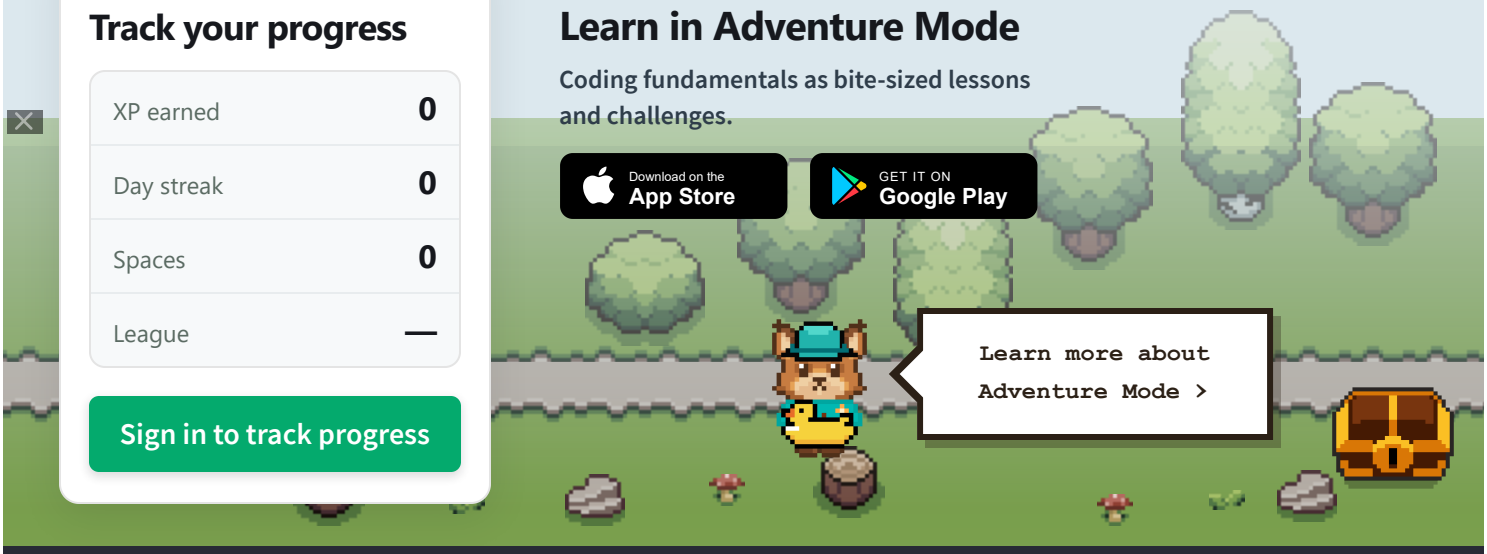
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